

Multi-sensor dataset of a tendon-driven continuum robot in dynamic motion

Andrea Gotelli^{1*}, Tongjia Zheng¹, Angela Peloso², Spencer Teetaert¹,
Puspita Triana Dewi¹, Greg Zetko¹, Chengnan Shentu¹, Jessica Burgner-Kahrs¹

¹University of Toronto Robotics Institute, University of Toronto, Canada

²Department of Electronics, Information and Bioengineering, Politecnico di Milano, Italy

September 16, 2026

Abstract

We present a multi-sensor dataset of a single-segment tendon-driven continuum robot undergoing quasi-static motions, dynamic trajectories, and contact interactions with a tracked object in the workspace. Six sensing modalities are recorded synchronously: optical motion capture of five disk poses along the backbone, fiber-optic distributed strain sensing (with shape reconstruction), tendon tension and displacement, base force–torque, and an external object pose and contact force. The dataset is organized into three subsets targeting distinct research tasks: quasi-static configurations for mechanical parameter identification and static model validation; dynamic trajectories (planar sweeps, star patterns, circular paths, and Lissajous curves at two actuation speeds) for dynamics modeling, state estimation, and learning-based methods; and dynamic motions with simultaneous contact measurements for contact interaction analysis. Both raw sensor recordings and processed data (low-pass filtered at 15 Hz, uniformly resampled at 100 Hz, and time-aligned) are provided in a consistent file structure with documented variable naming. Dataset quality is characterized through cross-modal validation, including pairwise cross-correlation of all sensor streams and force–kinematics consistency checks. The robot platform is assembled from commercially available components with a fully documented design.

Background & Summary

Continuum robots are a class of flexible manipulators whose slender, elastic bodies continuously deform along their entire length, in contrast to conventional rigid-link robots that articulate at discrete joints. Inspired by biological structures such as elephant trunks and octopus arms, they are actuated by tendons, pneumatics, or other distributed mechanisms, and are well suited to navigation in confined or unstructured environments [1, 2]. Early work established static models based on constant-curvature kinematics and Cosserat rod theory [3, 4]. As the field matured, dynamic modeling received increasing attention [5, 6, 7, 8]. **4.1, 5.1:** More recently, continuum robots have exposed to shape or force estimation [9, 10], control via the Clarke transform [11] and machine learning methods applied to both statics [12] and dynamics [13, 14, 15]. Each of these is developed and validated on its own custom platform, illustrating the same lack of a shared, reusable dataset that motivates this work. Parallel efforts have studied contact and environmental interaction [16].

Despite this active research landscape, experimental validation in continuum robotics typically relies on custom prototypes that are instrumented with lab-specific sensors under conditions that are difficult to reproduce externally. While experimental data is occasionally released as supplementary material, it is usually scoped to a specific model or control method, making it difficult to reuse across different frameworks or to draw systematic comparisons. As a result, the field lacks curated, multi-modal datasets that support systematic comparison and reproducibility. Existing public datasets, such as [17], provide valuable kinematic benchmarks for a concentric tube continuum robot, but do not include force measurements, dynamic motion regimes, or contact interactions: the quantities needed to benchmark dynamics models, study force–kinematics consistency, and analyse contact behaviour in tendon-driven systems. This limited availability of open, multi-modal datasets creates a bottleneck for the field: it lim-

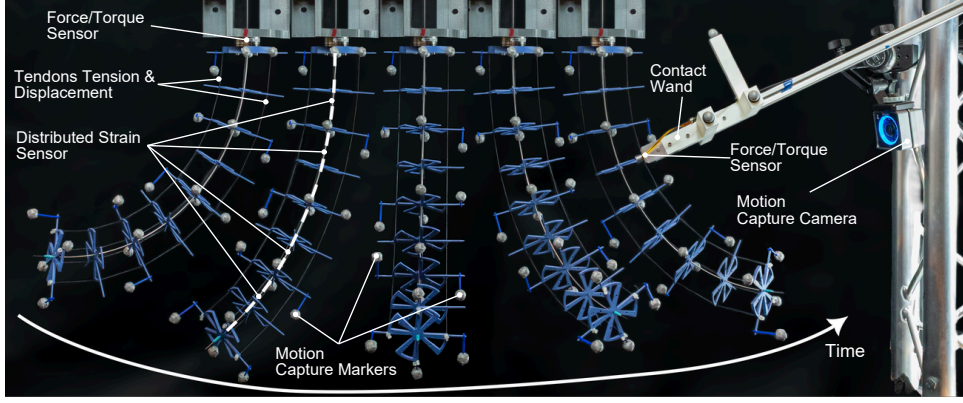


Figure 1: Snapshots of the continuum robot prototype overlaid across representative configurations, with annotations identifying the six sensing modalities integrated in the platform: optical motion capture frames distributed along the backbone, fiber-optic shape sensing, tendon tension and displacement measurements, and base and contact force-torque measurements.

its reproducibility of results and hinders systematic benchmarking of competing models, state estimation frameworks and learning-based methods on common ground.

The contrast with adjacent robotics fields is instructive. In autonomous driving, KITTI [18] unified evaluation of perception, odometry, and object detection enabling, for the first time, direct and reproducible comparison of algorithms on common ground. In robotic grasping, the Dex-Net dataset [19] combined synthetic depth images with analytic grasp quality labels at scale, catalyzing a generation of data-driven grasp planners that could be trained and benchmarked consistently. In aerial robotics, the EuRoC MAV dataset [20] provided stereo-visual and inertial sequences, becoming the de facto standard for evaluating visual-inertial odometry and SLAM algorithms. In each case, a shared, well-documented dataset provided a common experimental ground that accelerated reproducible benchmarking across competing methods. Continuum robotics lacks an equivalent: no common experimental reference exists against which dynamics models, state estimators, or learning-based controllers can be evaluated and compared.

2.1: In this context, the challenge is that continuum robots cannot be benchmarked the way KITTI, Dex-Net, or EuRoC MAV benchmark their domains: their behavior is not solely a function of a control policy applied to a generic body, but is inherently coupled to the mechanical design and material properties of the platform itself, an instance of embodied intelligence in which part of the system’s dynamic behavior is encoded directly in its hardware. A curated, richly instrumented dataset can still serve as a common baseline for comparing modeling, state estimation, and ~~control approaches~~ under identical hardware and environmental conditions. It also gives labs that already operate their own tendon-driven platforms a reference case study to benchmark against. *I don’t know how this dataset can be used for control. Drop “control” unless you have an idea how control can be validated on this dataset.*

To address this gap, we present a multi-sensor dataset of a single-segment tendon-driven continuum robot (TDCR) undergoing quasi-static and dynamic motions, including scenarios with contact in the workspace. Tendon-driven actuation is the most widely studied paradigm in continuum robotics, underpinning a large share of the modeling, control, and learning contributions in the literature, while other modalities such as pneumatic and concentric-tube designs address complementary application domains. Although limited to one single-segment robot in a pre-defined mounting condition, this dataset provides documented and synchronized measurements from six sensing modalities, covering the most relevant kinematic and force states that are measurable by available commercial sensors (Fig. 1). In detail, these sensing modalities are: motion capture of disk poses, distributed strain sensing (bending curvature and angle), tendon tension and displacement, base wrench, and contact forces. Both raw sensor recordings and processed data (filtered, time-aligned, and uniformly resampled) are provided, together with detailed documentation of file structure and variables. Collectively, these measurements span a broad range of state variables commonly considered in the continuum robotics literature. The dataset is organized into three subsets:

Subset 1 — Quasi-static motions provides steady-state configurations at known tendon tensions and motor displacements. It is suited for *mechanical parameter identification* and static model validation.

Subset 2 — Dynamic motions covers five trajectory types executed at two actuation speeds, pro-

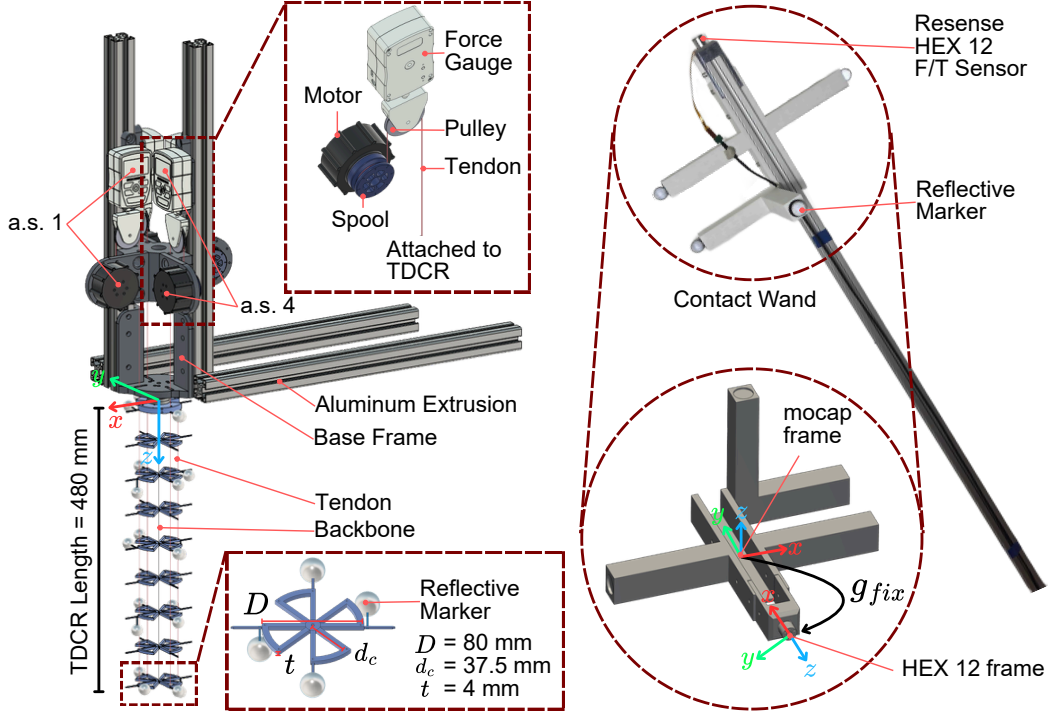


Figure 2: On the left, the design of the Tendon-Driven Continuum Robot (*a.s.* for actuator–sensor pair). On the right the contact wand with a representation of the motion capture (mocap) and force/torque sensor (HEX 12) frame with their relative homogeneous transformation g_{fix} .

ducing a range of dynamic regimes. It targets validation of *dynamics modelling* and *state estimation* research, including *learning-based* methods.

Subset 3 — Dynamic motions with contact extends Subset 2 with simultaneous contact force, contact location, and full robot state recordings, providing the quantities required to study contact modeling, force estimation, and state estimation under contact conditions.

To ensure data quality and internal consistency, cross-validation procedures are performed across sensing modalities, including comparisons between independent kinematic measurements and [1.3](#), [5.9: dynamics-model-based](#) consistency checks between heterogeneous force [2.3: measurements](#); sensor noise is additionally characterized from dedicated reference recordings. To facilitate reuse, all data (raw and processed) is provided in a standardized structure with consistent variable naming and detailed documentation. Processed data has uniform resampling and documented processing steps. The robot is assembled from commercially available components with 3D printed parts and a fully documented design, enabling other groups to replicate the platform and, possibly, extend the dataset.

Methods

This section describes the methodology used to produce the dataset. We first introduce the experimental platform and its mechanical design (Section Continuum Robot Platform), followed by a description of the six sensing modalities and their specifications (Section Sensors) and the instrumented contact wand used in subset 3 (Section Data collection accessory). We then describe the session preparation protocol (Section Session preparation), the recording sequences that constitute the three experimental subsets (Section Recording protocol), as well as the data acquisition pipeline, synchronization strategy and post-processing steps applied to produce the released data (Section Data Collection).

Table 1: **5.5:** Geometrical, material, and mass parameters of the experimental platform.

<i>NiTi backbone</i>			<i>Tendons</i>		
Length	ℓ	480 mm	Number	N_c	4 (2 antagonistic pairs)
Outer diameter	d_e	2.019 mm	Radial offset	d_c	37.5 mm
Inner diameter	d_i	1.487 mm	Spool radius	r_s	20 mm
5.6 :Mass	w_r	5.9 g	Material	Polyethylene, braided	
Young modulus (nomial value)	E	50 GPa	Specifications	12 lb, 0.12 mm	
<i>Spacer disks</i>					
Count		9			
Spacing		60 mm			
Outer diameter	D	80 mm			
Thickness	t	4 mm			
5.6 :Mass (tip disk)	w_t	8.90 g			
5.6 :Mass (base disk)	w_b	41.22 g			
5.6 :Mass (inter. disks)	w_d	5.25 g			
5.6 :Mass markers set (per disk)	w_t	1.70 g			

Continuum Robot Platform

The setup consists of a cantilever platform and a single-segment tendon-driven continuum robot (Fig. 2). The cantilever platform provides support to the continuum robot and is made of a 3D-printed base (PETG), four aluminum extrusion frames (40 mm T-slot profiles), and custom mounting adapters for the motors and force gauges. **5.3**The vertical and horizontal frames are rigidly joined using T-slot connectors, forming a stiff, leveled support structure. Its mounting stiffness is quantified in Section Mounting stiffness. This design aims to **5.11:** reduce sources of systematic error such structural compliance, and geometric uncertainty, ensuring precise and reproducible measurement of kinematic and force quantities across experiments. The continuum robot consists of a slender NiTi tubular backbone mounted with nine equally spaced disks at 6 cm intervals, material and geometric properties of the backbone, disks, and tendons are reported in Table 1. The backbone accommodates distributed internal strain sensing, while each disk is instrumented with a unique markers configuration enabling full six-degree-of-freedom pose reconstruction within the motion capture system. Four tendons are attached at the distal disk and routed through the intermediary disks with a constant radial offset of $d_c = 37.5$ mm, enabling controlled planar and spatial deformations. Each tendon is independently actuated by a dedicated motor–gauge assembly mounted at the base: tendons are redirected solely through a pulley integrated in the inline force gauge, with no additional pulleys between the robot and the actuator. This direct routing ensures that measured tensions correspond exactly to the loads applied to the robot, without friction losses at intermediate contacts. However, it also imposes a geometric constraint: the disk diameter must be large enough to accommodate the tendon routing at the required radial offset. To keep the robot design compact, the four motor–gauge assemblies are distributed symmetrically around the robot centerline, minimizing the required disk size to $D = 80$ mm, while preserving the symmetric loading condition. The four actuator–sensor pairs are labeled 1–4, corresponding to tendons aligned with the x , y , $-x$, and $-y$ directions **1.1:** (pairs 1 and 4, corresponding to the tendons aligned with x and $-y$ directions are labeled in Figure 2). The interface between the base platform and the robot base disk is designed to rigidly host a six-axis force–torque sensor, with the mounting reinforced to minimize compliance and isolate the measured boundary wrench from structural deformations of the frame.

1.2: Together, these design choices are intended so that each recorded quantity maps onto a specific **state** term used in standard tendon-driven continuum-robot models, such as the Cosserat-rod and Geometric Variable Strain (GVS) formulations [3, 4, 7]: the tendon tensions act as the actuation input, the base wrench provides the boundary condition, and the disk poses and reconstructed backbone shape describe the robot’s configuration. The direct tendon routing with no intermediate pulleys and the base F/T sensor placed at the exact mechanical boundary of the robot avoid the additional steps that would otherwise be needed to relate the sensor readings to these quantities. The symmetric arrangement of the four actuator–sensor pairs about the robot centerline further ensures that the actuation loads are balanced.

List a few more models, such as PCC, to avoid the confusion that the design is mainly for the Cosserat rod family of models. Some quantities may not directly map onto a model state and needs conversion. e.g., FBGS maps directly onto a PCC model but not a Cosserat rod model.

Sensors

The robot is instrumented with six sensing modalities covering both kinematic and force quantities. Table 2 summarizes the key specifications of each sensor; the subsections below describe what each modality measures, its output coordinate frame, accuracy, sampling rate, and calibration status.

Tendon tension

5.12 All four tendons are routed over pulleys to measure their tension (see Fig. 2). To **5.11**: reduce friction, each pulley is mounted on a bearing and supported by a custom bracket. This bracket is attached directly to the sensing axis of a M3-5 digital force gauge (Mark-10, New York, USA), providing a scalar tension measurement along the tendon axis. In this configuration, the force gauge measure the reaction force at the pulley which, for a 180 deg tendon routing, correspond to twice the actual tendon tension. **5.12**: To preserve these requirements, the pulleys have been 3D printed so their diameters cover exactly the distance between each tendon spools and the corresponding guiding hole in the robot base. The bore-to-bearing clearance is small enough that the press fit does not measurably deform the printed pulley, and roundness/concentricity was verified on a cylindrical test bench prior to assembly. The full-scale capacity is 22.2 N (5 lbf), with a resolution of 0.02 N and a manufacturer-specified accuracy of $\pm 0.3\%$ of full scale. Measurements are recorded directly in newtons. All four units were independently calibrated by an ISO 17025 accredited laboratory, with an uncertainty of ± 0.031 N. Data are acquired at an effective rate of approximately 335 Hz per gauge, limited by the serial request-response protocol round-trip latency.

Distributed strain sensing and shape reconstruction

Distributed strain sensing along the backbone is provided using a Fiber Bragg Grating (FBG) sensor with four-core fiber (FBGS International, Belgium) with 26 draw-tower gratings (DTGs) at 20 mm spacing, for an active sensing length of 500 mm. The fiber is interrogated by an FBG-Scan 904 spectrometer. The raw data is processed by the manufacturer’s proprietary software (ILLumiSense Shape CORE v1.1), which performs peak detection on the acquired spectra and reconstructs the bending curvature and angle (at the 26 equidistant points). These values are then integrated, by the proprietary software, to reconstruct the three-dimensional shape along the fiber (at 1 mm arc-length resolution). Both bending angle, axis and reconstructed shape are available at 100 Hz. These quantities are expressed in the fiber’s internal coordinate frame, with the arc-length parameter evolving along the fiber’s x -axis, as detailed in [21].

Base wrench

The six-axis wrench that the robot applies at its supporting base is measured using a mini40 force/torque sensor (ATI Industrial Automation, Apex, NC, USA), mounted between the base platform and the robot base disk. The sensor frame is aligned with the robot base frame (see Fig. 2). The sensing ranges are ± 40 N in F_x and F_y , ± 120 N in F_z , and ± 2 N m in T_x , T_y , and T_z . The measurement uncertainty at 95% confidence is ± 0.50 N (F_x , F_y), ± 0.90 N (F_z), ± 25 mN m (T_x), and ± 30 mN m (T_y , T_z), as reported in the factory calibration certificate. **5.33**: The transducer signals are digitized by a National Instruments USB-6210 Data Acquisition Device (DAQ) at 1000 Hz; every five consecutive samples are averaged before applying the manufacturer-provided calibration matrix, resulting in an effective frequency of 200 Hz. The timestamp is recorded immediately after this 5-sample (5 ms) acquisition window and is then corrected by subtracting half the window duration (2.5 ms), yielding a timestamp centered on the averaging window.

Rigid body poses

4.4: The six-degree-of-freedom pose (*i.e.* position and orientation) of selected objects in the workspace (e.g., the routing disks and the robot base) is measured using an optical motion capture system (OptiTrack, NaturalPoint, USA). The system tracks the position of passive retro-reflective markers (6 mm diameter spheres) arranged in unique configurations on rigid bodies; each configuration uniquely defines a frame in the workspace. Five such frames are attached to equally-spaced disks at arc-length positions $s \in \{0, 12, 24, 36, 48\}$ cm, providing discrete samples of the backbone pose. An additional frame, mounted on the contact object, is tracked in subset 3. The system is composed of five PrimeX13 and two PrimeX13W cameras arranged around the robot at a working distance from 1 m to 2 m, providing

Table 2: Summary of the sensor suite. Effective sampling rates account for communication overhead and software processing. FS = Full Scale, res. = resolution, cal. unc. = calibrated uncertainty.

Model / Manufacturer	Qty	Measured quantity	Range	Accuracy / Resolution	Eff. rate	Sub.
M3-5 (Mark-10, USA)	$\times 4$	Tendon force (1-axis)	0 N to 22.2 N	$\pm 0.3\%$ FS; res. 0.02 N; cal. unc. ± 0.031 N	~ 335 Hz	1–3
4-core fiber (FBGS Int., Belgium) +FBG-Scan 904	1 fiber, 26 DTGs	Curvature mm^{-1} , bending angle rad, 3-D shape mm	500 mm active fiber length	curvature and bending angle: 26 DTGs (≈ 20 mm spacing) shape: 502 points (≈ 1 mm spacing)	100 Hz	1–3
ATI mini40 (ATI Ind. Autom., USA) + NI USB 6210	1	6-axis wrench (F_x, F_y, F_z , T_x, T_y, T_z) at robot base	$F_{x,y}$: ± 40 N; F_z : ± 120 N; $T_{x,y,z}$: ± 2 N m	cal. unc. ± 0.50 N ($F_{x,y}$); cal. unc. ± 0.90 N (F_z); cal. unc. ± 25 mN m (T_x); cal. unc. ± 30 mN m ($T_{y,z}$)	~ 200 Hz	1–3
OptiTrack (NaturalPoint, USA)	PrimeX (13 $\times 5$, 13W $\times 2$) 6 rigid bodies	6-DoF pose of 5 disk frames + contact wand	$2 \times 2 \times 2$ m workspace	± 0.3 mm (position); ± 0.28 deg (Orientation)	120 Hz	1–3
Cybergear (Xiaomi, China)	$\times 4$	Shaft angle, tendon displacement (derived); velocity, torque	2π rad	14-bit encoder (16 384 counts/rev); angular res. $\approx 0.022^\circ$;	460 Hz	1–3
HEX 12 (Resense GmbH, Germany) + EVAL 100S-06-1	1	6-axis wrench (F_x, F_y, F_z , T_x, T_y, T_z) at contact point	$F_{x,y,z}$: ± 25 N; $T_{x,y,z}$: ± 125 mN m	1% FS; 3% crosstalk; 300% overload	100 Hz	3

markers positional accuracies of ± 0.15 mm and ± 0.30 mm respectively. Marker positions are processed and 3D rigid body poses solved by Motive 3.4 at 120 Hz, and output as position (x, y, z) and unit quaternion (q_x, q_y, q_z, q_w) using the scalar-last convention) in the motion capture ground frame. Mapping the markers position uncertainties into the frame pose uncertainty we obtain ± 0.3 mm for positions and ± 0.28 deg for orientations (roll, pitch, yaw).

Tendon displacement

The displacement of each tendon relative to the straight configuration is derived from the shaft angle measured by the encoder integrated in each actuator. Four Cybergear quasi-direct-drive motors (Xiaomi, China) are used, each fitted with a 14-bit single-turn absolute encoder reporting shaft angle at an effective feedback rate of approximately 460 Hz. The corresponding displacement is derived geometrically as $\Delta l = r_s \theta$, where $r_s = 20$ mm is the spool winding radius and θ is the shaft angle relative to the straight configuration. The encoder resolution is 14-bit over a single turn, corresponding to an angular resolution of $\approx 0.022^\circ$ and a tendon displacement resolution of ≈ 7.6 μ m. No additional calibration is applied beyond the encoder’s factory characterization. The home position, which defines the straight configuration reference, is re-initialized at each session startup. The tendon displacement is a geometric estimate derived from motor shaft angle and assumes a constant spool winding radius; any tendon stretch is not captured. Each motor additionally reports shaft velocity and output torque at every control cycle; these quantities are included in the raw data but have not been independently validated.

Contact wrench

Contact forces between the robot and objects in the workspace are measured using an HEX 12 six-axis force/torque sensor (Resense GmbH, Klingenberg, Germany), used exclusively in subset 3. The nominal measurement ranges are ± 25 N in F_x, F_y, F_z , and ± 125 mN m in T_x, T_y, T_z , with a manufacturer-specified accuracy of 1% of full scale and a crosstalk of 3%. The sensor is paired with the Resense miniature evaluation electronics board (EVAL 100S-06-1), which applies the manufacturer-provided calibration matrix onboard, outputting calibrated wrench components at 100 Hz. No additional in-house calibration was performed beyond the manufacturer-provided matrix; however, the sensor is tared immediately before each experiment to remove any static offset accumulated since power-on.

Data collection accessory

To collect data in subset 3, the robot must be brought into contact with an instrumented object whose interaction force and pose are simultaneously known, without occluding the disk markers from the motion capture cameras. The contact wand addresses this: it consists of a slender aluminum extrusion frame (20 mm T-slot profile) long enough for the operator to interact with the robot from outside the camera field of view, equipped with a Resense HEX 12 force/torque sensor at its tip and a motion capture frame on its body (Fig. 2).

Since the force/torque sensor and the motion capture [1.1: frames](#) are rigidly attached [1.1: to the wand body](#) but physically offset, a fixed transformation $g_{\text{fix}} \in SE(3)$ [5.7: \(\$g_{\text{fix}} = \langle R_{\text{fix}}, r_{\text{fix}} \rangle\$, with \$R_{\text{fix}} \in SO\(3\)\$ and \$r_{\text{fix}} \in \mathbf{R}^3\$ \)](#) is required to express the measured wrench in any frame other than the sensor frame. This transformation was defined during design, verified in assembly and is applied in post-processing. The rotational part accounts for the physical misalignment between the sensor connector orientation and the motion capture frame axes, and is composed of a rotation of 90° about the x -axis followed by a rotation of 30° about the z -axis. The translational offset $\mathbf{r}_{\text{fix}} = (0, -0.1137, 0)^T$ m corresponds to the distance along the wand body between the motion capture frame origin and the Resense sensor centre (see Fig. 2). Combined with the wand pose $g_{\text{wand}}(t)$ recorded by the motion capture system, the $g_{\text{contact}} = g_{\text{wand}} g_{\text{fix}}$ transform enables the contact wrench to be expressed in any desired reference frame; its use is demonstrated in the contact force validation (Section Technical Validation).

Session preparation

The robot is mounted on a cantilever platform with its tip pointing downward, *i.e.* in the direction of gravity (see Fig. 2). By mounting downward, the effect of gravity creates a stable equilibrium for the robot in a straight configuration, leading to more consistent motions and well-defined loading condition across all experiments. The same layout is used across all subsets of the dataset. To ensure consistency across all sensor modalities, every recording session follows a fixed preparation protocol. [5.14: This](#)

protocol removes three independent sources of session-to-session variability: drift in the motion-capture calibration, misalignment between the FBG fiber and the robot’s bending plane, and bias in the base force/torque sensor; each step below addresses one of these before data collection begins.

The OptiTrack camera system was calibrated in the lab using the manufacturer-provided wand calibration procedure. Once calibrated, the cameras perform continuous self-calibration during operation, maintaining tracking accuracy without requiring recalibration between sessions. Timestamps are assigned on the acquisition machine using the system wall clock. Raw poses are expressed in the global motion capture frame; the robot base-relative re-expression is applied in postprocessing.

The FBG sensor fiber is placed with its first grating coincident with the robot base, thus having 20 mm of sensing length extending from the robot tip. At the start of each data collection session, the FBG sensor fiber calibration matrix is updated twice to **5.11: reduce** measurement uncertainty. **5.14: The two updates remove two independent sources of error: a translational offset in the zero-strain reference, and a rotational offset between the fiber’s deformation plane and the robot’s bending plane.** First, using a self-leveling laser (Bosch Professional GLL50-20), the robot is verified to be straight along both the x and y axes, with tendon pre-tensions compensating any residual backbone deformation; the calibration matrix is then updated to match this straight reference configuration. Second, the robot is bent significantly in the yz plane by actuating the $+y$ tendon, with the laser confirming that the deformation remains planar and correcting any out-of-plane component via the remaining tendons; the calibration matrix is updated again from this configuration to eliminate any residual torsional offset in the fiber measurements.

Before each experiment, the base frame is verified to be level using a circular bubble level, ensuring that the gravitational load direction is consistent with the most common modeling assumptions for tendon-driven continuum robots. **5.14: Unlike the FBG calibration above, which is performed once per session, level and straightness are re-verified before every individual experiment, since small perturbations to the base or backbone can accumulate over the course of a session.** The laser is then used to confirm that the robot is straight. In this configuration, the tendons are pre-tensioned to approximately 1 N each, with a maximum difference between the two tendons of an antagonistic pair <0.2 N. This step **5.11: reduces** the risk of tendon slack, a well-known source of modeling error in tendon-driven continuum robots [22].

The mini40 data acquisition module applies an initial bias each time it is powered on. Therefore, at the beginning of each recording, the sensor is tared using the current reading. As a result, the measured F_z contains an unknown constant offset (*i.e.*, it does not represent the robot’s true internal axial force). This offset may be approximated as the robot’s total weight (101.27 gram) minus the pretension applied to the tendons.

Recording protocol

This section describes the recordings that constitute the dataset. Once the session preparation procedures described in the previous section are complete, two reference recordings are performed before the main data collection begins. The dataset is then organised into three subsets corresponding to distinct experimental conditions, each targeting a different class of robot behaviour.

Reference recordings. Two preliminary recordings are performed, stored in the `references/` folder. First, with all tendons released, a baseline recording is acquired to characterize the noise floor of all sensors in the absence of any load or motion (`released.config`). Then, with the robot verified straight using the Bosch laser, a second recording is acquired to provide an estimate of the residual alignment error in the FBG sensor and motion capture measurements under the known straight configuration (`straight.config`).

Quasi-static subset. The 16 recordings in `quasi-static/` are obtained by driving one pair of antagonist tendons, with maximum angular velocity of $\omega_{\max} = 0.3$ rad/s, so that inertial and damping effects are negligible and the robot reaches a sequence of near-equilibrium configurations **5.24: , i.e. the tendon tension is balanced by the robot elasticity.** Bending is applied independently along $+x$, $-x$, $+y$, and $-y$, each at four commanded amplitude levels corresponding to maximum motor angles of 100, 150, 180, and 200°. **5.24: To apply the desired angle, the motor command file has been manually modified for each recording.** Each recording sweeps the bending amplitude in small increments, producing synchronized measurements of tendon tensions, encoder angles, OptiTrack disk poses, and FBG curvatures across a range of near-equilibrium configurations.

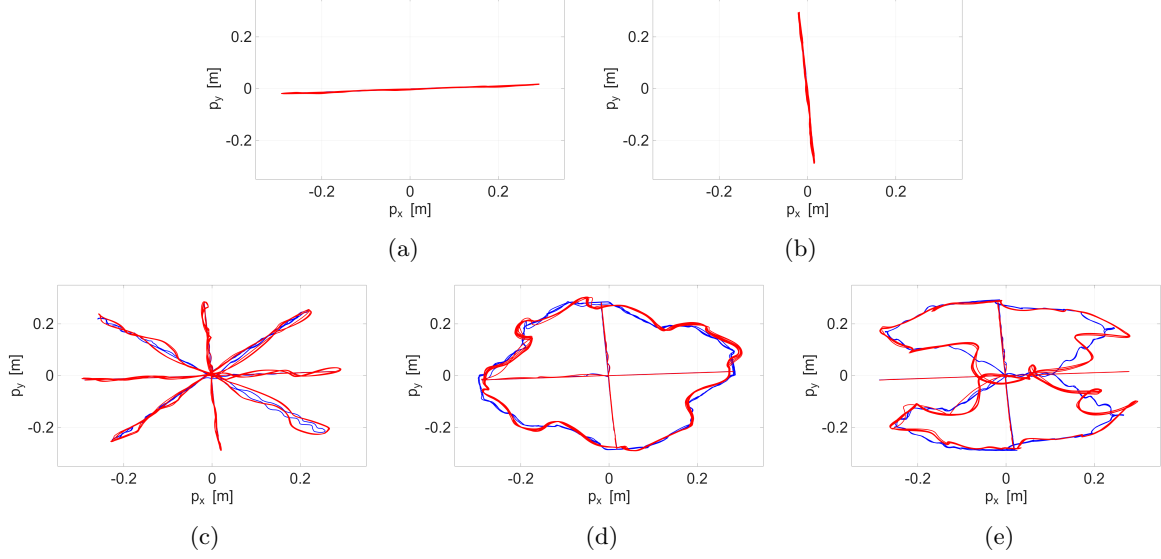


Figure 3: Motion of the robot tip (xy plane) for the planar motions along x (a) and y (b) planes, the star (c), circular (d), and Lissajous (e) trajectories. Blue: slow speed ($\omega_{\max} = 1$ rad/s); red: fast speed ($\omega_{\max} = 3$ rad/s). **3.2:** The initial x - z and y - z planar excursions visible on the circular and Lissajous trajectories are calibration sweeps, executed before the target motion to register the FBG sensor and OptiTrack frames.

Dynamic motion subset. The 10 recordings in `dynamic_motion/` cover five trajectory types executed at two speeds ($\omega_{\max} = 1$ rad/s and $\omega_{\max} = 3$ rad/s), producing a range of dynamic regimes (Fig. 3). The slow speed produces moderate dynamic effects while the fast speed amplifies inertial and damping contributions. The planar trajectories: (`plane_x` and `plane_y`), drive a single antagonist tendon pair confining the motion to either the xz - or yz -plane, respectively. These recordings serve as controlled baselines: the actuation involves one degree of freedom and the expected robot shape is a planar arc. The `star` trajectory alternates between the x - and y -tendon pairs and their combination in successive segments, sweeping the tip through four quadrants in a star-like pattern, introducing transitions between bending planes and richer coupling between the two actuation channels.

Finally, the `circle` and `Lissajous` trajectories drive both tendon pairs simultaneously to produce out-of-plane tip motion with sustained coupling between the x and y channels, executed as a sequence of discrete target configurations under closed-loop position control rather than continuous sinusoidal commands. **5.23:** The `Lissajous` waypoints are generated from the parametric equations $x(t) = R\sin(at + \delta)$, $y(t) = R\sin(bt)$ (sixteen waypoints per cycle; $a = 2$, $b = 1$, $\delta = 0$, $R = 180^\circ$); the `circle` trajectory uses eight hand-tuned waypoints approximating the same nominal shape ($a = b = 1$, $\delta = \pi/2$), with small adjustments compensating tendons slack asymmetrically between cardinal and diagonal points. The Lissajous trajectory additionally changes direction multiple times per cycle, generating more varied accelerations and richer spectral content than the circle.

For every non-planar trajectory (*i.e.* star, circular and Lissajous curves) the robot executes a calibration routine before the target motion. The routine consists of a planar sweep in the x - z plane (positive x followed by negative x), then a planar sweep in the y - z plane (negative y followed by positive y). These calibration sweeps facilitates the alignment of the FBG sensor and the OptiTrack frames, in post-processing, before the non-planar data is interpreted (Section Post-processing).

Contact motion subset. The four recordings in `contact_motion/` extend the dataset with external contact scenarios. All experiments use the instrumented contact wand (Section Data collection accessory), so that the applied contact force vector and the 6-DoF wand pose are recorded alongside all other robot sensors. Contact experiments are restricted to the planar- y trajectory (with $\omega_{\max} = 1$ rad/s) because the geometrical constraints of the experimental setup — the thin spacer disks, the OptiTrack marker brackets, and the small tip area of the Resense sensor — make reliable and repeatable contact difficult to achieve on more complex three-dimensional motions. `push_retract` records the response of

the un-actuated robot to an external tip load: the tendons are fully released and the wand is slowly pushed against the tip then retracted. `plane_y_contact_midway` is an actuated experiment in which the robot executes the trajectory while the wand is held fixed against the spacer disk located 18 cm from the base, blocking the robot’s motion at that point and redistributing internal forces. `plane_y_poke_y` and `plane_y_poke_x` are dynamic perturbation experiments in which, during the robot motion, the wand delivers brief transverse pokes in the y - and x -directions respectively, exciting in-plane and out-of-plane oscillations.

Data Collection

This section describes the acquisition pipeline, synchronization strategy, and post-processing steps applied to produce the released dataset.

Data collection interfaces

Data acquisition is handled by a combination of manufacturer-provided software and custom scripts, depending on the sensor. Two sensors rely on dedicated proprietary software running as independent processes. The FBG sensor is managed by the manufacturer’s Shape CORE software, which converts detected spectral peak into bending curvature and angle used to reconstruct the 3-D shape; the output is streamed over a local TCP connection to a custom C++ client that timestamps and buffers each frame. The OptiTrack motion capture system uses Motive 3.4, running on a dedicated machine, which solves 3-D rigid-body poses and streams them over an UDP protocol (unicast) using the NatNet 4.4 SDK. A custom Python client on the acquisition machine receives and timestamps each frame. **5.14:** No reconstruction step is needed for the remaining sensors, whose native output (a force reading, an encoder angle) is already the quantity of interest, so they are read directly by dedicated Python processes. The M3-5 force gauges connect directly via USB using a serial request–response protocol (one process per gauge). The others each require a hardware interface between the PC and the sensor output: the mini 40 force/torque sensor uses a National Instruments USB 6210 DAQ card, the Cybergear motors use a USB–CAN adapter to access the CAN 2.0 bus, and the HEX 12 contact force sensor uses a EVAL 100S-06-1 evaluation interface. Each acquisition process runs on an independent thread and logs data to a dedicated CSV file upon recording completion. **5.31:** Timestamps are assigned via `time.time()` in Python and epoch-anchored `std::chrono::system_clock` in C++. The acquisition machine runs Python 3.13, on which Windows’ `time.time()` has microsecond-level timestamp resolution. **5.34:** The motor process is launched last and waits 3 s before sending position commands. This delay, chosen manually, ensures all sensors are already streaming and capture a pre-motion baseline at the start of each recording. **5.32:** The four motor states stored under this single timestamp are read sequentially over the CAN bus, so a small readout latency separates the true sampling instants of the four motors; this latency is quantified in Section Temporal synchronization. All acquisition scripts are released alongside the dataset.

Synchronization

To **5.11:** reduce experimental complexity, sensors are connected directly to the acquisition machine without external hardware triggering. Each acquisition process runs on an independent thread, sharing the host’s OS clock to ensure timestamp alignment within the limits of OS scheduling jitter. **5.25:** This shared clock rules out drift between streams; it does not guarantee that all timestamps mark the same stage of measurement, since each sensor’s timestamp is assigned at a different point in its own acquisition process (e.g. a DAQ block read, a serial response, a sequential multi-axis query). Two sensors are an exception: the OptiTrack system runs Motive on a dedicated host PC and streams poses over a network link, and the FBG sensor output passes through the manufacturer’s proprietary reconstruction software; both introduce an additional pipeline delay. These delays are measured empirically and their impact on temporal alignment is discussed in Section Temporal synchronization; required corrections are addressed in postprocessing (Section Post-processing). Residual timing offsets should be considered when analyzing high-frequency transients.

Post-processing

Post-processing aims to align all sensor measurements to a common reference frame and time base, to reduce measurement noise, and to provide uniformly sampled signals for downstream analysis. **5.14:** Each of these three goals responds to a different mismatch introduced by acquiring every sensor through

Table 3: Complete file and column documentation for each experiment folder. Subscript $n \in \{1, 2, 3, 4\}$ denotes motor index (motors 1 & 3 form the x -plane pair; 2 & 4 the y -plane pair). Subscript $k \in \{0, \dots, 4\}$ denotes disk index from base to tip. Raw files include a header row; processed files are header-less.

File	Column(s)	Unit	Description
Raw Files			
dataMotor.csv	timestamp	s	CPU wall-clock
	phase_index	–	Trajectory phase (0=idle; successive integers = motion segments)
	home $_n$ ($n = 1-4$)	bool	Homing completed flag, motor n
	target $_n$ _rad	rad	Commanded shaft angle, motor n
	abs_angle $_n$ _rad	rad	Absolute encoder reading, motor n
	rel_angle $_n$ _rad	rad	Relative angle from home, motor n
	velocity $_n$ _rad/s	rad/s	Shaft angular velocity, motor n
	torque $_n$ _Nm	N m	Output torque, motor n
dataMark10-*.csv (4 files: +x, -x, +y, -y)	timestamp	s	CPU wall-clock
	tension (N)	N	Reaction force at pulley
dataATIFT.csv	timestamp	s	CPU wall-clock
	Fx (N), Fy (N), Fz (N)	N	Base reaction force
	Tx (Nm), Ty (Nm), Tz (Nm)	N m	Base reaction torque
dataOptiTrack.csv ($k = 0-4$ for robot disks $k = 5$ for the contact wand)	timestamp.s	s	CPU wall-clock
	disk. k .frame_number	–	Motive software frame counter
	disk. k .is_valid	bool	Tracking validity (0 = occluded)
	disk. k .x/y/z	m	3-D position of disk k
	disk. k .q[xyzw]	–	Orientation quaternion,
dataFBGS.csv	Timestamp	s	CPU wall-clock
	Curvature_0–_25 (26)	1/mm	Bending curvature
	Angle_0–_25 (26)	rad	Bending angle
	Shape_0–_1505 (1 506)	mm	Flattened $[x, y, z]$ triplets of the reconstructed 3-D shape
dataResenseFT.csv (subset 3 only)	timestamp (s)	s	CPU wall-clock
	Fx, Fy, Fz	N	Contact force
	Tx, Ty, Tz	mN m	Contact torque
Processed Files (15 Hz low-pass filtered, time aligned and resampled at 100 Hz)			
angles.csv	col 1: time	s	Time
	cols 2–5: $\theta_1, \theta_2, \theta_3, \theta_4$	rad	Motor shaft rotation angles
tendon_tensions.csv	col 1: time	s	Time
	cols 2–5: $T_{+x}, T_{+y}, T_{-x}, T_{-y}$	N	Tendon tensions
base_wrench.csv	col 1: time	s	Time
	cols 2–4: F_x, F_y, F_z	N	Base reaction force
	cols 5–7: T_x, T_y, T_z	N m	Base reaction torque
mocap_frames.csv	col 1: time	s	Time
	5.17: $roll_k - pitch_k - yaw_k$	rad	Disk orientation (w.r.t. robot base frame)
	5.17: x_k, y_k, z_k ($k = 0-4$)	m	Disk position (w.r.t. robot base frame)
fbgs_shapes.csv	col 1: time	s	Time
	cols 2–1507: $[x_i, y_i, z_i]$, 5.16: $i = 1-502$	m	Reconstructed 3-D shape
fbgs_strains.csv	col 1: time	s	Time
	cols 2–27: k	1/m	Bending curvature
	cols 28–53: θ	rad	Bending angle
5.20: wrench_wand.csv (subset 3 only)	col 1: time	s	Time
	cols 2–4: F_x, F_y, F_z	N	Contact force
	cols 5–7: T_x, T_y, T_z	N m	Contact torque
wand_pose.csv (subset 3 only)	col 1: time	s	Time
	x, y, z	m	Contact wand position (w.r.t. robot base frame)
	$roll-pitch-yaw$	rad	Contact wand orientation (w.r.t. robot base frame)

an independent process: raw recordings differ in coordinate frame, in absolute timing reference, and in native sampling rate. In the following, we detail all steps addressing each mismatch in turn, their implementation is in `process_data.m` (MATLAB), released alongside the dataset. Both raw and processed files are provided; users who require different filtering, resampling, or frame conventions should work from the raw files.

Coordinate frame alignment. **5.14:** Because OptiTrack and the FBG sensor reports poses or shapes in their own frame, kinematic measurements must first be brought into a single, physically meaningful frame before they can be interpreted as robot deformation. Raw motion capture poses are expressed in the Motive global frame, which is arbitrary and calibration-dependent. They are re-expressed relative to a body frame coincident with the base of the robot (see Fig. 2), making poses changes directly interpretable as robot deformation. **5.22:** This base-frame definition is recomputed independently for every recording, from a 120 samples-window. A second correction addresses static misalignment introduced by assembly tolerances and marker placement uncertainty. Since the robot is known to be straight in the `references/straight_config/` recording, the deviation of each disk pose from the expected straight configuration in that recording is a rigid-body offset attributable entirely to sensor placement error rather than robot deformation. These per-disk correction transforms are computed once from the reference recording and applied in the local frame of each disk at every timestep, removing the static offset from all subsequent pose measurements.

5.14: The FBG sensor is subject to the same kind of frame ambiguity, in its own coordinates. In this case, the Shape CORE software reconstructs the fiber shape with the arc-length parameter evolving along the x -axis of its local fiber frame. Thus, a rotation is therefore applied to map the local fiber frame to the robot body frame. Any residual rotational offset between the fiber and the robot bending plane is additionally corrected via a singular-value decomposition performed on the tip trajectory during the initial planar calibration sweep: the principal directions of the tip motion identify the bending plane and the corresponding rotation angle θ_{FBG} . A matching SVD is applied to the motion-capture tip trajectory, yielding θ_{mocap} , and the correction applied to the reconstructed shape is a rotation of $\theta_z = \theta_{\text{FBG}} - \theta_{\text{mocap}}$ about the z -axis.

Tendon tension correction. Each Mark-10 force gauge is mounted on the axle of a redirecting pulley around which the tendon wraps 180° . As result, a factor-of-two correction is applied to all four tension measurements in postprocessing; while the raw measurements retain the uncorrected gauge readings.

5.14: Unlike the frame and timing corrections above, this correction is a fixed, known bias determined entirely by the sensor’s mounting geometry, so the same constant factor is applied to every recording without further calibration.

Temporal alignment. **5.14:** Because each acquisition process assigns its own timestamps with no hardware trigger linking them, the recordings share a common clock but not a common reference instant, which the following step establishes. All timestamps are expressed relative to the first motor command so that $t = 0$ marks the start of motion. Samples recorded before this event (sensor warm-up, static baseline) are retained in the raw files and discarded from the processed files. **5.29:** Additionally, a fixed offset of 12.25 ms is subtracted from all FBG sensor timestamps to compensate for the pipeline delay introduced by the proprietary reconstruction software. This value corresponds to the pipeline delay estimated empirically via cross-correlation against OptiTrack (Section Temporal synchronization); subtracting this offset indicates that FBG timestamps lag the common OS clock.

Low-pass filtering. **5.14:** With every signal now expressed in a common frame and time base, the remaining disagreement between sensors is measurement noise, which the following filtering step reduces. Each signal is filtered with a zero-phase, fourth-order Butterworth low-pass filter (cutoff 15 Hz, applied via `filtfilt`). A Butterworth filter was chosen for its maximally flat passband, which avoids amplitude distortion below the cutoff frequency. The cutoff frequency was selected empirically using the OptiTrack tip position, which provides the highest-bandwidth kinematic signal in the dataset. The robot tip trajectory was filtered at a candidate cutoff f_c and again at $2f_c$; the cutoff was increased until the root mean square error between the two filtered signals fell below 10^{-4} m, indicating that the signal carries negligible power above f_c . Zero-phase filtering via `filtfilt` ensures that no phase distortion is introduced. Users interested in higher-frequency content should work from the raw files. **5.19:** Note that the sampling frequency used to design each signal’s filter is estimated from the mean inter-sample

interval of that signal’s timestamps. Subsequently, the signal is resampled onto an uniform time grid, spanning the same duration and sample count; this avoids bias from irregular timestamp spacing.

Resampling. **5.14:** Finally, to provide synchronized measurements, all filtered signals are resampled onto a uniform 100 Hz grid by linear interpolation. This rate matches the native rate of the slowest sensors (FBG and HEX 12) and provides more than twofold Nyquist margin relative to the 15 Hz filter cutoff. Linear interpolation is sufficient because all signals are already band-limited by the preceding filter.

Data Records

The dataset is hosted on figshare [23]. It is organised in four top-level directories corresponding to the three experimental subsets and the reference recordings introduced in Section Background & Summary. Table 3 documents every file type, its columns, units, and content.

File naming and outputs

5.39: The dataset includes per-sensor recordings saved directly from each acquisition process (referred to as “raw” data, meaning unmodified by the authors’ own post-processing pipeline), as well as post-processed versions that are filtered, time-aligned, and resampled. For the FBG sensor, this “raw” file (`dataFBGS.csv`) already contains curvature, bending angle, and reconstructed 3-D shape produced by the interrogator’s proprietary Shape CORE software. Each sensor process writes one CSV file per experiment following the naming convention `data{Sensor}.csv` (e.g. `dataMotor.csv`, `dataMark10_x.csv`, `dataATIFT.csv`, `dataOptiTrack.csv`, `dataFBGS.csv`). Column headers and units for each file are listed in Table 3. The post-processed data are saved in a `processed/` subfolder within each experiment directory, containing one file per quantity, all resampled to 100 Hz with a shared time column. **5.40:** The curvature and bending angle at each of the 26 gratings, computed internally by ShapeCore during shape reconstruction, are converted from the interrogator’s native units (curvature: mm^{-1} to m^{-1}), time-aligned, low-pass filtered, and resampled at 100 Hz. The reconstructed 3-D shape is also re-expressed in the robot body frame (as in Section Post-processing) **5.20:** For subset 3, the `processed/` folder additionally contains `wrench_wand.csv` and `wand_pose.csv`; the corresponding raw file is `dataResenseFT.csv`. Both raw and processed data are released.

Dataset Structure

The files described above are organized in a hierarchical directory tree. At the top level, four directories separate the three experimental subsets and the reference recordings:

```
data/
  quasi_static/
    static_bend_x_100/      ... static_bend_x_200/
    static_bend_x_n100/    ... static_bend_x_n200/
    static_bend_y_100/      ... static_bend_y_200/
    static_bend_y_n100/    ... static_bend_y_n200/
  dynamic_motion/
    plane_x_slow/   plane_x_fast/
    plane_y_slow/   plane_y_fast/
    star_slow/      star_fast/
    circle_slow/    circle_fast/
    Lissajous_slow/ Lissajous_fast/
  contact_motion/
    push_retract/      plane_y_contact_midway/
    plane_y_poke_y/     plane_y_poke_x/
  references/
    released_config/
    straight_config/
```

Each experiment folder (e.g., `circle_slow/`) contains the raw sensor CSV files at the top level and a `processed/` subfolder with the filtered, resampled data:

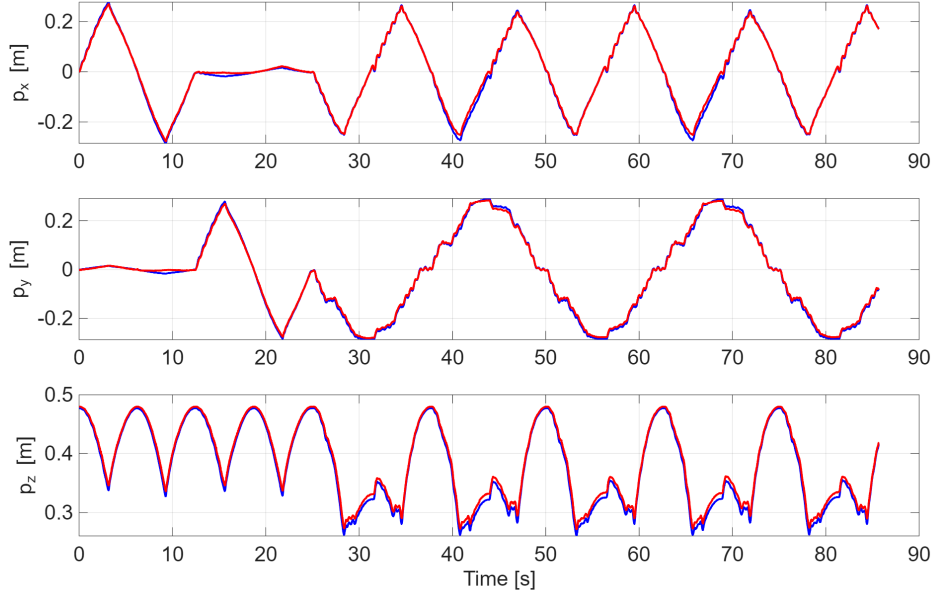


Figure 4: Trajectory of the robot tip as measured by the OptiTrack motion capture system (blue) and the FBG sensor (red) for the Lissajous (slow) trajectory.

```
<experiment>/
  dataMotor.csv          dataATIFT.csv
  dataMark10_+x.csv      dataMark10_-x.csv
  dataMark10_+y.csv      dataMark10_-y.csv
  dataFBGS.csv           dataOptiTrack.csv
  processed/
    angles.csv           tendon_tensions.csv
    base_wrench.csv      mocap_frames.csv
    fbgs_shapes.csv      fbgs_strains.csv
    wrench_wand.csv      wand_pose.csv      (subset 3 only)
```

The `quasi_static/` subset contains 16 experiments corresponding to planar bends along $\pm x$ and $\pm y$ at four commanded amplitudes (maximum tendon actuation angles of 100, 150, 180, and 200°). The `dynamic_motion/` subset contains 10 experiments (5 trajectories $\times$ 2 speeds). The `contact_motion/` subset contains 4 experiments: `push_retract` (quasi-static, no actuation); and three dynamic recordings at slow speed (`plane_y_contact_midway`, `plane_y_poke_y`, `plane_y_poke_x`). The `references/` folder stores two baseline recordings: `released_config` (tendons released, robot at rest) and `straight_config` (robot held straight with laser alignment).

Technical Validation

This section presents the analyses performed to verify the technical quality and internal consistency of the recorded dataset. The dataset contains two families of measurements: kinematic quantities (OptiTrack poses, FBG sensor strain and reconstructed shape, actuator encoder angle) and force quantities (M3-5 cable tensions, mini40 base wrench, HEX 12 contact force). Within each family, sensors may be directly compared. Kinematic sensors are cross-validated by comparing OptiTrack tip positions with FBG sensor-reconstructed shapes, and encoder-derived tendon displacements with motion-capture-based geometric reconstructions. On the other hand, force measurements cannot be compared directly without a physical model. Therefore, a reduced-order dynamics model is employed as a cross-correlation tool to verify consistency, as detailed in Section Consistency of tendon tension measurements.

Table 4: Sensor noise characterisation. *Measured std* is the standard deviation computed over the reference recording with the robot at rest, characterising the random noise floor.

Sensor	Quantity	Unit	Measured std
M3-5 (+x)	Tendon tension	N	0.02 [†]
M3-5 (+y)	Tendon tension	N	0.02 [†]
M3-5 (-x)	Tendon tension	N	0.02 [†]
M3-5 (-y)	Tendon tension	N	0.02 [†]
Actuator (+x)	Joint angle	rad	2.0×10^{-4}
Actuator (+y)	Joint angle	rad	2.0×10^{-4}
Actuator (-x)	Joint angle	rad	2.0×10^{-4}
Actuator (-y)	Joint angle	rad	2.0×10^{-4}
mini40	F_x	N	0.013
	F_y	N	0.017
	F_z	N	0.944
	T_x	N m	7.1×10^{-4}
	T_y	N m	4.3×10^{-4}
	T_z	N m	7.9×10^{-3}
HEX 12	F_x	N	0.011
	F_y	N	0.0076
	F_z	N	0.013
	T_x	N m	5.1×10^{-5}
	T_y	N m	2.9×10^{-5}
	T_z	N m	2.4×10^{-5}
OptiTrack disk 0	Position (x, y, z)	mm	(0.005, 0.005, 0.003)
	Orientation (X, Y, Z)	deg	(0.0043, 0.0034, 0.0049)
OptiTrack disk 1	Position (x, y, z)	mm	(0.005, 0.005, 0.004)
	Orientation (X, Y, Z)	deg	(0.0022, 0.0029, 0.0032)
OptiTrack disk 2	Position (x, y, z)	mm	(0.007, 0.008, 0.008)
	Orientation (X, Y, Z)	deg	(0.0049, 0.0126, 0.0071)
OptiTrack disk 3	Position (x, y, z)	mm	(0.008, 0.008, 0.003)
	Orientation (X, Y, Z)	deg	(0.0034, 0.0032, 0.0049)
OptiTrack disk 4	Position (x, y, z)	mm	(0.011, 0.013, 0.005)
	Orientation (X, Y, Z)	deg	(0.0068, 0.0109, 0.0077)
FBG sensor	Curvature noise along arc	m ⁻¹	0.011
	Bending angles noise along arc	rad	0.663
	Shape noise along arc (x, y, z)	mm	(0.002, 0.194, 0.233)
	Tip position (x, y, z)	mm	(0.006, 0.476, 0.508)

[†]Manufacturer-specified gauge resolution, since all four Mark-10 units read identically zero at null load (tendons released).

Sensor noise baseline

A dedicated static recording (**references/released_config**) was collected with the robot hanging at rest and all tendons released. This recording provides a sensor-independent characterisation of the noise floor, decoupled from any robot motion. Table 4 reports the standard deviation of each sensor measured over the full reference recording. For the FBG sensor, we note that the tip position, curvature, and bending angle are outputs of Shape CORE’s proprietary nonlinear reconstruction pipeline; their noise is state-dependent and not guaranteed to be Gaussian. The bending angle values in particular should be interpreted as conservative upper bounds: in the near-straight configuration the robot lies close to a parameterization singularity where any bending direction produces an indistinguishable straight shape, inflating the measured standard deviation; this effect diminishes as curvature increases.

Mounting stiffness

5.3: The frame-to-base mounting stiffness was quantified in a dedicated bench test: while logging the base ATI force/torque sensor, a series of manual pushes was applied to the robot base and the resulting relative displacement between markers on the base disk and on the support frame was tracked with the

Table 5: RMSE of tip position between OptiTrack and FBG sensor for each dynamic trajectory. Values are in millimeters; percentages in parentheses are relative to the motion range.

Trajectory	RMSE x [mm] (%)	RMSE y [mm] (%)	RMSE z [mm] (%)
plane_x slow	8.38 (1.5%)	N/A	4.08 (2.8%)
plane_x fast	9.13 (1.6%)	N/A	4.96 (3.1%)
plane_y slow	N/A	8.88 (1.6%)	3.92 (2.7%)
plane_y fast	N/A	8.99 (1.5%)	4.55 (2.9%)
star slow	6.89 (1.2%)	7.85 (1.4%)	5.32 (1.9%)
star fast	7.82 (1.3%)	8.06 (1.4%)	5.79 (1.9%)
circle slow	10.43 (1.8%)	9.22 (1.6%)	7.12 (4.1%)
circle fast	10.22 (1.7%)	9.49 (1.6%)	7.38 (3.8%)
Lissajous slow	8.83 (1.6%)	8.95 (1.5%)	6.61 (3.0%)
Lissajous fast	9.58 (1.6%)	9.07 (1.6%)	6.77 (2.8%)

OptiTrack system. Across five pushes (25 N to 48 N) spanning different directions, the measured stiffness ranged from 56 N mm^{-1} to 316 N mm^{-1} , confirming the mount is directionally anisotropic. Even the most compliant direction is more than sufficient for the loads the robot itself generates: the in-plane forces and torques transmitted through this connection during the recorded trajectories peak at 0.4 N to 0.8 N and 0.15 N m, corresponding to deflections below 0.01 mm. The largest base reaction, along z , is transmitted directly by the vertical frames to the motors.

Kinematic cross-validation

Comparison of recorded shape with FBG sensor and Motion Capture

5.28: Every disk pose carries a per-sample validity flag set by Motive (`disk_k_is_valid` in Table 3, 0 indicating an occluded or invalid frame); we checked this flag across the entire released dataset and found it valid throughout, so no frames were excluded from the comparisons below. **5.27:** OptiTrack and FBG sensors provide two independent sensing modalities for reconstructing the robot’s shape. The FBG sensor’s base frame is aligned to the OptiTrack one using the planar calibration sweeps performed at the start of each recording (Section Recording protocol), the following therefore assesses the cross-modal shape consistency. We compare the tip position recorded by OptiTrack with the tip position reconstructed from the FBG sensor, evaluated at the same arc-length coordinate. Table 5 reports the root-mean-square error (RMSE) for all dynamic trajectories. The tip RMSE remains below 11 mm in the primary motion axes and below 8 mm in the vertical (z) direction across all trajectories and speeds, an error being 1.2–4.1% of the range of motion. The source of this error can be attributed to the dead-reckoning error in the shape reconstruction [21], as well as placement uncertainties of the OptiTrack frames and FBG fiber. **5.27:** Nonetheless, these results confirm that the FBG-reconstructed shape remains consistent with the OptiTrack-tracked shape after alignment, across all dynamic trajectories tested. For a qualitative assessment, Figure 4 shows the overlay of these two signals for the Lissajous trajectory.

4.2: Similar comparison is performed for the contact recordings comparing the per-disk RMSE against a representative non-contact dynamic recording (`dynamic_motion/Lissajous_fast`). Table ?? reports these values at each of the five tracked disks. The RMSE during contact is at or below the level observed in the non-contact recording at every disk and axis, indicating no measurable increase in mocap-FBGS discrepancy attributable to contact-induced local deformation. In this case, since the intermediate disks move very little, absolute RMSE are reported to provide results comparable across recordings regardless of motion range.

fix referencing

Validation of tendon Displacement via Motion Capture

An independent geometric estimate of the tendon displacement is derived from the motion-capture data to cross-validate the encoder-based measurements recorded by the actuators (see Section Sensors). To this aim, the pose of the five measured disk frames are interpolated to locate the intermediate disks (not provided with OptiTrack frames). Positions are interpolated with a shape-preserving piecewise-cubic Hermite (PCHIP) scheme, while orientations are interpolated by converting each rotation matrix to a unit quaternion, applying PCHIP independently to each quaternion component, and normalizing the result to the unit sphere. The consistency of the quaternion hemisphere is enforced between successive

Table 6: **4.2:** RMSE between OptiTrack and FBG-reconstructed disk position for the contact-motion recordings, with `Lissajous fast` as a non-contact reference. Values are absolute, in millimeters. Disk index k follows the base-to-tip convention ($k = 4$: tip).

Disk k	Recording	RMSE x [mm]	RMSE y [mm]	RMSE z [mm]
0	<code>plane_y_contact_midway</code>	0.099	0.078	0.054
	<code>plane_y_poke_x</code>	0.092	0.123	0.591
	<code>plane_y_poke_y</code>	0.045	0.110	0.111
	<code>push_retract</code>	0.124	0.117	0.089
	<code>Lissajous fast (non-contact)</code>	0.150	0.223	0.042
1	<code>plane_y_contact_midway</code>	1.386	2.849	0.750
	<code>plane_y_poke_x</code>	1.178	2.588	0.261
	<code>plane_y_poke_y</code>	0.852	2.634	0.748
	<code>push_retract</code>	1.450	2.316	1.013
	<code>Lissajous fast (non-contact)</code>	2.345	2.589	0.702
2	<code>plane_y_contact_midway</code>	2.738	5.751	1.058
	<code>plane_y_poke_x</code>	2.349	5.778	1.271
	<code>plane_y_poke_y</code>	1.838	5.863	1.099
	<code>push_retract</code>	2.640	4.662	1.009
	<code>Lissajous fast (non-contact)</code>	5.354	5.454	1.477
3	<code>plane_y_contact_midway</code>	4.598	7.984	2.489
	<code>plane_y_poke_x</code>	4.294	7.952	2.942
	<code>plane_y_poke_y</code>	3.914	8.056	2.559
	<code>push_retract</code>	3.770	7.090	0.912
	<code>Lissajous fast (non-contact)</code>	7.754	7.616	3.933
4	<code>plane_y_contact_midway</code>	7.086	9.443	4.617
	<code>plane_y_poke_x</code>	6.796	9.071	5.117
	<code>plane_y_poke_y</code>	6.680	9.256	4.694
	<code>push_retract</code>	4.488	9.629	1.071
	<code>Lissajous fast (non-contact)</code>	9.579	9.073	6.766

disks to avoid sign-flip artifacts. In each interpolated frame, the absolute position of every tendon is computed by applying the local tendon offset d_c on each disk pose. The total length of the tendon ℓ_c^d , where the suffix d indicates the relation with the disk pose, is then approximated as the piecewise-linear sum of Euclidean distances between consecutive attachment points along the interpolated backbone. Tendon displacement is defined as the change in length relative to the straight (pre-motion) configuration, estimated by averaging the first $N_{\text{ref}} = 10$ frames:

$$\Delta \ell_c^d(t) = \ell_c^d(t) - \bar{\ell}_c^{d, \text{ref}}, \quad \bar{\ell}_c^{d, \text{ref}} = \frac{1}{N_{\text{ref}}} \sum_{k=1}^{N_{\text{ref}}} \ell_c^d(t_k). \quad (1)$$

Table 7 reports the RMSE between encoder-based and motion-capture-based tendon displacement for all dynamic trajectories. The RMSE remains below 5.6 mm across all trajectories, corresponding to less than 4.8% of the motion range, confirming that the encoder recordings and the motion-capture poses are geometrically consistent. Agreement between them constitutes a geometric consistency check, confirming that the encoder records and the motion-capture poses are mutually consistent and that the assumed tendon-routing geometry is correct.

Consistency of contact measurements

The contact subset involves two quantities that require independent verification: the position of the contact sensor and the contact wrench it measures. These are first checked in isolation and then in combination.

The contact wand carries a dedicated OptiTrack frame. To verify that the sensor position is correctly recorded throughout the robot’s deformation range, the Euclidean distance between the tracked wand position and the centre of the robot tip disk was computed over the full `push_retract` recording. The

Table 7: RMSE of tendon displacement between encoder and OptiTrack reconstruction for each dynamic trajectory. Values are in millimeters; percentages in parentheses are relative to the tendon motion range. Tendons are labeled by direction ($+x$, $+y$, $-x$, $-y$).

Trajectory	$+x$ [mm] (%)	$+y$ [mm] (%)	$-x$ [mm] (%)	$-y$ [mm] (%)
plane_x slow	2.11 (1.9%)	N/A	3.98 (3.5%)	N/A
plane_x fast	2.89 (2.6%)	N/A	4.01 (3.5%)	N/A
plane_y slow	N/A	2.93 (2.6%)	N/A	2.40 (2.1%)
plane_y fast	N/A	2.75 (2.4%)	N/A	3.00 (2.6%)
star slow	2.50 (2.2%)	3.29 (2.9%)	2.68 (2.4%)	2.10 (1.8%)
star fast	2.97 (2.6%)	3.83 (3.4%)	2.73 (2.4%)	2.29 (2.0%)
circle slow	3.96 (3.4%)	4.93 (4.2%)	3.21 (2.8%)	3.30 (2.8%)
circle fast	4.55 (3.9%)	5.57 (4.8%)	3.51 (3.0%)	3.26 (2.8%)
Lissajous slow	2.86 (2.6%)	3.26 (2.8%)	2.64 (2.3%)	2.25 (1.9%)
Lissajous fast	3.28 (2.9%)	3.44 (3.0%)	2.70 (2.3%)	2.31 (2.0%)

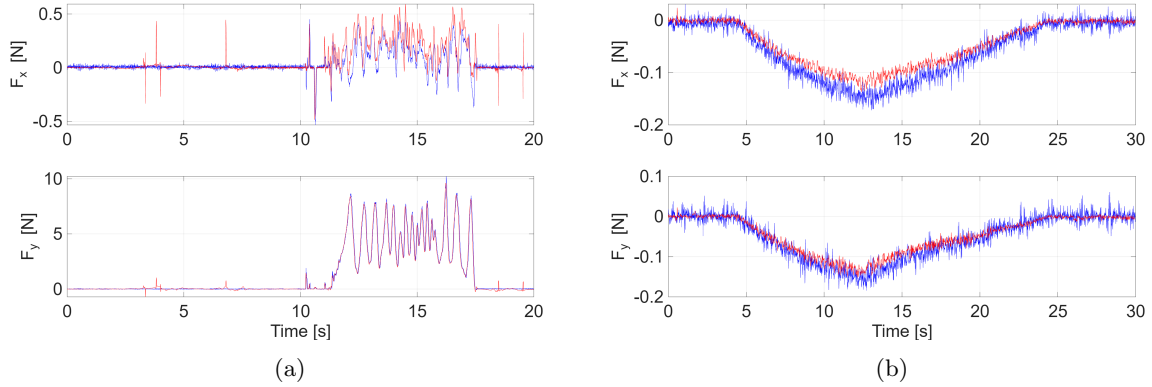


Figure 5: Comparison of the measured forces (F_x and F_y) and torques (T_x and T_y) from the measurement of the HEX 12 (red) to the mini40 (blue) sensors for the touch_base (a) and push_retract (b) recordings. **3.2:** Both sensors were tared immediately before each recording, so each trace begins from a zero baseline.

mean distance was ≈ 38 mm with negligible standard deviation, consistent with the known physical offset between the Resense sensor origin and the disk centre and confirming accurate position tracking across all configurations.

The wrench measurements from the HEX 12 contact sensor were validated by comparing them against the base wrench, measured by the mini40, via a rigid-body transformation. While the full wrench is recorded and reported, our validation focuses explicitly on the primary contact force axis, which represents the most critical metric for our interaction setup. Two tests were conducted, both with all tendons released so that the only external load acting on the robot was the contact force applied by the wand. In the first test, the wand was pressed against the robot base disk along the y -axis of its frame, keeping the lever arm between the contact point and the mini40 sensing origin short and well-defined. The HEX 12 wrench was mapped into the mini40 coordinate frame via the adjoint map

$$\mathcal{W}_{\text{contact}}^{\text{base}} = -\text{Ad}_{g_{\text{base},\text{contact}}} \mathcal{W}_{\text{Resense}}^{\text{wand}}, \quad (2)$$

where $g_{\text{base},\text{contact}} = g_{\text{base}}^{-1} g_{\text{contact}}$ is the OptiTrack-measured relative pose between the base disk and the wand, and the minus sign accounts for the action-reaction principle. The adjoint transports a wrench between frames without assuming rigidity of the connecting body, so Eq. (2) remains valid as the continuum robot deforms. Figure 5a shows close agreement between the transformed Resense wrench and the ATI-F/T measurement, confirming that the Resense calibration and the geometric transformation are mutually consistent. **5.4:** Table 8 reports RMSE, mean bias, maximum absolute error, and Pearson correlation r for samples where the wand is in contact, identified geometrically from the mocap-tracked distance between the wand and the surface it presses on. Bias is the mean signed error over these N contact samples, $\frac{1}{N} \sum_{i=1}^N (\mathcal{W}_{\text{contact},i}^{\text{base}} - \mathcal{W}_{\text{mini40},i}^{\text{base}})$, where $\mathcal{W}_{\text{mini40}}^{\text{base}}$ is the wrench measured directly by the mini40 at the base. The primary contact axis, F_y , agrees closely (RMSE 0.172 N, bias -0.043 N, $r =$

Table 8: **5.4:** Quantitative agreement between the Ad_g -transported HEX 12 wrench and the mini40 measurements, restricted to contact window (determined by Euclidean distance to contact object).

Recording	Axis	RMSE [N]	Bias [N]	Max. abs. err. [N]	r
touch_base	F_x	0.151	-0.113	0.335	0.525
	F_y	0.172	-0.043	0.539	0.998
push_retract	F_x	0.019	+0.017	0.038	0.997
	F_y	0.012	+0.009	0.027	0.998

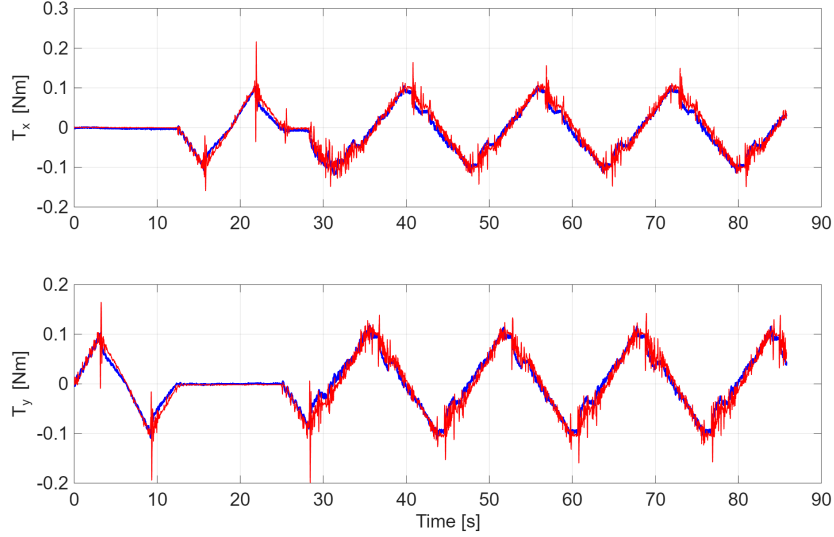


Figure 6: Comparison of the simulated torque at the base (red) with the measured torque by the (unfiltered) mini40 sensor (blue) using the circular motion.

0.998); the off-axis F_x component is weaker ($r = 0.525$) and small in absolute terms (with RMSE 0.151 N), as expected since the wand was pressed along y . **5.20:** Because the applied contact force is small, this comparison applies an additional 2 Hz low-pass filter on top of the pipeline’s 15 Hz filter (Section Post-processing) to both the base and contact wrench signals, avoiding RMSE inflation from passband noise.

In the second test (using the `push_retract` recording), the wand was used to push the robot tip through a quasi-static deformation, combining both checks: the position tracking established above feeds the full pose chain from wand to base (lever arm up to 0.48 m) into the same transformation. Figure 5b shows good agreement in the force components across the full deformation range. **5.4: Restricted to contact samples,** HEX 12 force components agree closely with the mini40 measurement (F_x : RMSE 0.019 N, $r = 0.997$; F_y : RMSE 0.012 N, $r = 0.998$; see Table 8). In both tests, the sensor’s vertical (z) axis carries no informative contact-force signal and is excluded from the comparison. The torque components exhibit an offset that grows with robot deformation; this is expected: as the robot bends, a growing fraction of its weight is registered by the HEX 12 but not at the mini40. The force agreement across the full deformation range confirms the consistency of both force sensors and the pose-based wrench transformation.

Consistency of tendon tension measurements

Consistency of tendon tension measurements can be verified by comparison with the base wrench. However, these measurements cannot be directly compared without a physical model relating tendon forces to base reactions. As an auxiliary consistency check, we employ a forward dynamics simulation based on the Geometric Variable Strain (GVS) approach [7], using the measured tendon tensions as input to predict the base torque and comparing the result against the mini40 measurements. This check is model-dependent: the agreement reflects the joint quality of the data and the accuracy of the characterised

Table 9: RMSE between GVS-simulated and mini40-measured base torque for each dynamic trajectory. Values are absolute (N m) with the percentage of the signal range in parentheses. The two components T_x and T_y correspond to the bending moments about the x and y axes at the robot base.

Trajectory	Speed	RMSE T_x [N m] (%)	RMSE T_y [N m] (%)
plane_x	slow	N/A	0.0162 (7.8%)
	fast	N/A	0.0474 (19.4%)
plane_y	slow	0.0160 (7.6%)	N/A
	fast	0.0417 (17.0%)	N/A
star	slow	0.0141 (6.4%)	0.0138 (6.2%)
	fast	0.0342 (13.9%)	0.0362 (15.2%)
circle	slow	0.0155 (6.9%)	0.0163 (7.2%)
	fast	0.0357 (13.7%)	0.0391 (15.2%)
Lissajous	slow	0.0148 (6.3%)	0.0147 (7.0%)
	fast	0.0389 (17.6%)	0.0387 (15.8%)

mechanical parameters, and should not be interpreted as an independent sensor validation. The GVS simulation code [5.10: has been released alongside the postprocessing code](#). The equivalent bending rigidity EI of the backbone was estimated from the quasi-static recordings; inertial parameters were derived from CAD without further identification.

Figure 6 compares the simulated and mini40-measured base torque for the circular trajectory; the overall trend is well captured, with residual discrepancies (e.g. torque spikes at fast speed) attributed to the approximate distributed inertia parameters. Table 9 reports the RMSE across all dynamic trajectories for the two base-torque components T_x and T_y ; the component not excited by a given planar trajectory (e.g. T_x for `plane_x`) is omitted. For slow trajectories, RMSE remains below 0.02 N m ($< 8\%$ of the signal range); at fast speed it increases to ~ 0.04 N m ($\sim 15\text{--}20\%$), consistent with the greater contribution of inertial terms that are only approximately characterized. The qualitative agreement across all trajectories supports the internal coherence of the force and kinematic recordings, noting that residual discrepancies reflect the approximate nature of the model parameters rather than inconsistencies in the data.

Temporal synchronization

All sensors share a single operating-system clock (see Section Synchronization), bounding the theoretical inter-stream error to OS scheduling jitter. As noted in Section Synchronization, two sensors are exceptions due to their reliance on proprietary software: Motive reconstructs rigid-body frames from marker positions, while Shape CORE processes the raw FBG output into bending curvature and angle, integrating these values to reconstruct the fiber shape. Their temporal alignment was verified empirically via pairwise normalized cross-correlation $r \in [-1, 1]$, where $|r| = 1$ indicates a perfectly linear relationship between two signals — positive for co-varying pairs, negative for kinematically anti-correlated ones — and $|r| \approx 0$ indicates insufficient shared motion content for reliable lag estimation. [5.26: Four fast trajectories \(`circle\_fast`, `Lissajous\_fast`, `plane\_x\_fast`, and `plane\_y\_fast`\) were selected because their high-amplitude, spectrally rich motion produces sharp cross-correlation peaks. For motor–sensor pairs, only the axis predominantly excited by each trajectory is used to avoid kinematic coupling artifacts on passive axes. The circular trajectory is reported as the primary reference because it simultaneously excites all three Cartesian axes, yielding the most complete and robust estimates; the remaining trajectories confirm the results on their respective dominant axes.](#)

[5.26:](#) The mocap–FBG lag Δ_{OF} was measured on all four fast dynamic trajectories (`circle_fast`, `Lissajous_fast`, `plane_x_fast`, `plane_y_fast`), giving a mean of -12.25 ms. As an internal consistency check, this lag can be triangulated against the motor–mocap (Δ_{MO}) and motor–FBG (Δ_{MF}) lags, since $\Delta_{MF} = \Delta_{MO} + \Delta_{OF}$ if all three are independently measured. This check is restricted to `circle_fast`: its circular motion excites the x and y bending axes in quadrature, so each tendon pair’s motor signal correlates with a distinct, identifiable mocap/FBG axis. In this case, $\Delta_{MO} = -2.54$ ms and $\Delta_{MF} = -15.64$ ms; with this trajectory’s own $\Delta_{OF} = -13.04$ ms, the predicted Δ_{MF} is -15.58 ms, matching the measured value to within 0.06 ms.

Indirect evidence of alignment is provided by the RMSE values in Tables 5 and 7: a residual timing offset would inflate RMSE in proportion to signal slope, yet tip-position RMSE between OptiTrack and FBG

sensor remains below 11 mm (1.8% of motion range) and tendon-displacement RMSE remains below 5.6 mm (4.8%), consistent with geometric and calibration uncertainties.

Cross-correlation analysis was deliberately omitted for the tendon tension and base force-torque sensors, as computing alignment for these streams is both impractical and unnecessary. First, because force and torque measurements are inherently coupled to system dynamics, their readings reflect a complex system response that does not correlate directly or linearly with kinematic measurements. **5.25:** Second, because these sensors interface directly with the acquisition machine rather than through Motive or Shape CORE, any additional latency is bounded by the duration of a single serial read, DAQ block, or CAN poll, rather than by an external pipeline delay. For the base wrench, this bound is the 2.5 ms DAQ-block correction applied to its timestamp (Section Sensors); for the motor encoders, it is the four-motor CAN readout latency, measured at 1.68 ms on average (std 0.09 ms), i.e. each individual motor query takes approximately 0.42 ms.

Usage Notes

Getting started. The dataset is available for download from figshare [23]. **5.22:** A README.md at the root of the code archive documents the raw-to-processed workflow end to end while DEPENDENCIES.md lists software and hardware dependencies for both the acquisition and post-processing stages (see also Section Data Collection). After downloading, users should select the experimental subset of interest (quasi-static, dynamic motion or contact) and, within it, the trajectory type they wish to analyze. All subsets expose the same file structure under their `processed/` subfolder, so downstream code does not need to distinguish between them: each experiment folder contains one CSV per sensor stream with a common time axis, pre-filtered signals, and aligned timestamps. Users who wish to regenerate the processed files from the raw CSVs, or to apply alternative filtering parameters, should run the postprocessing file (`process_data.m`) on the desired experiment folder. The `references/straight_config/` recording should be loaded first to compute the OptiTrack frame correction, while `references/released_config/` provides a static noise baseline for computing per-experiment sensor bias.

Recommended subset per task. The three subsets address complementary research tasks and can be used independently or in combination. Researchers working on mechanical parameter identification are recommended to use `subset_1` (quasi-static): each recording provides disk poses from OptiTrack alongside simultaneous tendon tensions and motor encoder readings, enabling estimation of backbone bending rigidity EI and tendon axial stiffness. Researchers working on state estimation, system identification, or benchmarking of modeling and prediction methods should use `subset_2` (dynamic), which provides full multi-sensor time series across a range of trajectories and actuation speeds. Researchers working on contact dynamics or wrench estimation should use `subset_3` with contact force and pose measurements. **4.7:** An example of using tendon tension and base wrench measurements together with a dynamics model is reported in Technical Validation.

Sensor-specific guidance.

FBG sensor shape data. The FBG fiber may exhibit a residual rotational offset with respect to the robot bending plane, which can be corrected in post-processing using the planar calibration sweeps (see Section Recording protocol). Users working with the raw measurements should apply this correction themselves. In addition, the shape reconstruction algorithm integrated in the proprietary FBGS acquisition software accumulates a dead-reckoning error along the fiber arc: small per-segment orientation errors integrate into a positional drift that grows towards the tip.

Measured base wrench. The mini40 exhibits a high noise floor on the F_z axis (≈ 0.94 N). Users should restrict analyses requiring sub-newton axial force resolution to F_x and F_y ; T_x and T_y are the most reliable quantities for bending moment analysis.

Motor encoder data. Shaft velocity and output torque are included in the raw measurements data but have not been independently validated; users relying on these quantities should treat them as indicative.

Software requirements. The raw and processed data are stored as plain CSV files and can be loaded in any programming language. The post-processing pipeline (`process_data.m`) was developed and tested in MATLAB R2025b, using only the built-in functions `butter`, `filtfilt`, and `interp1` from the Signal Processing Toolbox. **5.31:** The acquisition scripts require Python 3.13 or later: on Windows,

`time.time()` only reaches microsecond precision from this version onward; earlier Python versions fall back to the coarser ~ 15.6 ms-resolution system call and should not be used to reproduce the acquisition timing described in Section Synchronization. **5.22:** Moreover, the following libraries should be installed: `numpy`, `pyserial`, `python-can`, `nidaqmx`, `matplotlib`) together with vendor-supplied drivers/SDKs for the OptiTrack NatNet client, the National Instruments DAQ used by the ATI sensor, and the PEAK-System PCAN-USB adapter used by the tendon actuators; the FBG interrogator client is distributed as a pre-compiled executable, with its C++ source included for users who need to rebuild it. **5.22:** All file paths are relative to the code repository and are set once, in the `PATHS/SETTINGS` block at the top of `process_data.m`. Since the released code and data share the same relative layout as distributed on figshare (`code/` and `data/` as siblings), the pipeline runs unmodified once the repository is downloaded.

Limitations of applicability. This dataset is not suitable for all continuum robotics research tasks, and users should be aware of the following scope constraints before applying it. All processed signals are low-pass filtered at 15 Hz and resampled at 100 Hz; it is not appropriate to study dynamics or contact transients above 15 Hz. Raw data is provided for users who require higher-frequency content. The dataset covers a single robot platform with a fixed morphology, backbone material, and mounting configuration; results obtained on this platform may not generalize directly to robots with different backbone stiffness, tendon routing, or aspect ratios, and such a generalization should be validated separately. **2.1:** This constraint reflects the embodied, hardware-coupled nature of tendon-driven continuum robot dynamics discussed in Section Background & Summary. Contact scenarios are restricted to interactions with a rigid instrumented wand along the planar- y trajectory; soft, distributed, or multi-point contact is not represented. **5.36:** The processed signals are filtered with a zero-phase (`filtfilt`) Butterworth filter, which is acausal: each output sample depends on both past and future input samples. This exact filtering procedure cannot be reproduced in a causal, online or real-time setting. Users who require causal filtering should apply their own causal filter (~~e.g. a Kalman type filter~~), relevant in particular for online or state-estimation use of Subset 2. Kalman filters need a dynamical model to work and thus not applicable here. Saying causal filter is sufficient.

Generalization across nominally identical platforms. **3.1:** Model transferability across identical robots is limited by specific manufacturing and material variability: backbone viscoelastic creep and hysteresis, marker and sensor placement tolerances, and tendon pretension. A model trained on this dataset is not expected to generalize zero-shot to a different physical unit of the same design. The quasi-static subset, already used here to estimate the backbone’s equivalent bending rigidity EI (Section Consistency of tendon tension measurements), provides the same grounds for characterizing a new unit’s residual physics, i.e. the deviations from the nominal model a replica would otherwise carry over uncorrected. Cross-platform generalization is outside the scope of this data descriptor.

Reference recordings. The `released_config` recording was acquired with all tendons released and the robot hanging freely under gravity. Due to assembly tolerances in the alignment of the disks and residual plasticity accumulated in the NiTi backbone after repeated bending cycles, this configuration exhibits a small residual deflection rather than being perfectly straight. The `straight_config` recording addresses this: the robot is verified to be straight along both bending planes using a self-leveling laser before acquisition, ensuring a geometrically straight reference independent of assembly tolerances.

Motor encoder data. **5.23:** The motors are position-controlled with discrete waypoint commands for the `circle` and `Lissajous` trajectories (see Section Data Records for the governing equations). The commanded way-points also include a small (15%) reduction applied to the retracting tendon of each antagonist pair relative to the ideal geometric target, compensating for slack/pretension asymmetry between the push and pull sides.

Data 5.15: & Code Availability

The dataset described in this work is publicly available on figshare [23], **4.3:** license CC-BY 4.0. The repository contains all raw and processed sensor recordings organized as described in the Data Records section (Section Data Records).

All acquisition and processing scripts are released under the MIT License and are available as part of the dataset repository [23].

Author Contributions

A.G. conceptualized the dataset and data processing pipeline, performed the data collection and technical validation, and contributed to the sensor interface development, calibration, and contact wand design. T.Z. contributed to the sensor interface development, the actuator trajectory design, and the robot platform design and fabrication. A.P. contributed to the development of the acquisition software and the prototype design and fabrication. S.T. configured the motion capture system. P.D. designed and fabricated the robot platform. G.Z. designed the contact wand, performed the calibration and verification of the HEX 12 force/torque sensor, and contributed to the contact subset data collection and robot photography. C.S. contributed to the motor command interface and readout pipeline. J.B.-K. supervised the project. All authors contributed to writing and editing the manuscript.

Competing Interests

The authors declare no competing interests.

Funding

We acknowledge the support of the Natural Sciences and Engineering Research Council of Canada (NSERC, RGPIN-2025-04343), the Canada Foundation for Innovation (CFI), and the Ontario Research Fund (ORF). A.P. was supported by the ARTERY project funded by the EU Horizon 2020 research and innovation programme under grant agreement No. 101017140.

References

- [1] G. Robinson and J. Davies, “Continuum robots - a state of the art,” *International Conference on Robotics and Automation*, vol. 4, pp. 2849–2854, 1999.
- [2] J. Burgner-Kahrs, D. C. Rucker, and H. Choset, “Continuum robots for medical applications: A survey,” *IEEE Transactions on Robotics*, vol. 31, no. 6, pp. 1261–1280, 2015.
- [3] R. J. Webster and B. A. Jones, “Design and kinematic modeling of constant curvature continuum robots: A review,” *The International Journal of Robotics Research*, vol. 29, no. 13, pp. 1661–1683, 2010.
- [4] D. C. Rucker and R. J. Webster, “Computing jacobians and compliance matrices for externally loaded continuum robots,” *IEEE International Conference on Robotics and Automation*, pp. 945–950, 2011.
- [5] F. Renda, V. Cacucciolo, J. Dias, and L. Seneviratne, “Discrete cosserat approach for soft robot dynamics: A new piece-wise constant strain model with torsion and shears,” *International Conference on Intelligent Robots and Systems (IROS)*, pp. 5495–5502, 2016.
- [6] J. Till, V. Aloï, and C. Rucker, “Real-time dynamics of soft and continuum robots based on cosserat rod models,” *The International Journal of Robotics Research*, vol. 38, pp. 723–746, 2019.
- [7] F. Boyer, V. Lebastard, F. Candelier, and F. Renda, “Dynamics of continuum and soft robots: A strain parameterization based approach,” *IEEE Transactions on Robotics*, vol. 37, no. 3, pp. 847–863, 2021.
- [8] F. Renda, A. Mathew, and D. F. Talegon, “Dynamics and control of soft robots with implicit strain parametrization,” *IEEE Robotics and Automation Letters*, vol. 9, pp. 2782–2789, 2024.
- [9] G. Zhang, Z. Chen, and L. Wang, “Virtual-work based shape-force sensing for continuum instruments with tension-feedback actuation,” *arXiv:2501.05418*, 2025.
- [10] E. Wang, Z. Deng, C. Pan, B. He, and J. Zhang, “Estimating continuum robot shape under external loading using spatiotemporal neural networks,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. DOI: 10.1109/IROS60139.2025.11245975.

- [11] C. Muhmann, R. M. Grassmann, M. Bartholdt, and J. Burgner-Kahrs, “Toward dynamic control of tendon-driven continuum robots using Clarke transform,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. DOI: [10.1109/IROS60139.2025.11247518](https://doi.org/10.1109/IROS60139.2025.11247518).
- [12] M. Bensch, T.-D. Job, T.-L. Habich, T. Seel, and M. Schappler, “Physics-informed neural networks for continuum robots: Towards fast approximation of static cosserat rod theory,” *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 17293–17299, 2024.
- [13] Y. Liu, K. Luo, Q. Tian, and H. Hu, “An efficient data-driven framework of hybrid dynamics for real-time modeling and control of continuum robots,” *Mechanism and Machine Theory*, vol. 220, p. 106361, 2026. DOI: [10.1016/j.mechmachtheory.2026.106361](https://doi.org/10.1016/j.mechmachtheory.2026.106361).
- [14] Y. Du, L. Liu, and Z. Zhang, “A physics-informed neural network framework for real-time model predictive shape manipulation of deformable linear objects,” *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 1666–1682, 2026.
- [15] H. M. Hansen, B. K. Sæbø, K. Y. Pettersen, J. T. Gravdahl, and M. Di Castro, “Data-driven dynamic modeling of a tendon-actuated continuum robot,” *arXiv:2605.18720*, 2026. DOI: <https://doi.org/10.48550/arXiv.2605.18720>.
- [16] P. Rao, O. Salzman, and J. Burgner-Kahrs, “Towards contact-aided motion planning for tendon-driven continuum robots,” *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4687–4694, 2024.
- [17] R. M. Grassmann, R. Z. Chen, N. Liang, and J. Burgner-Kahrs, “A dataset and benchmark for learning the kinematics of concentric tube continuum robots,” *International Conference on Intelligent Robots and Systems (IROS)*, pp. 9550–9557, 2022.
- [18] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, “Vision meets robotics: The KITTI dataset,” *International Journal of Robotics Research*, vol. 32, no. 11, pp. 1231–1237, 2013.
- [19] J. Mahler, J. Liang, S. Niyaz, M. Pokorny, S. Niyaz, F. T. Pokorny, S. Niyaz, P. Abbeel, and K. Goldberg, “Dex-Net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics,” in *Robotics: Science and Systems*, 2017.
- [20] M. Burri, J. Nikolic, P. Gohl, T. Schneider, J. Rehder, S. Omari, M. W. Achtelik, and R. Siegwart, “The EuRoC micro aerial vehicle datasets,” *International Journal of Robotics Research*, vol. 35, no. 10, pp. 1157–1163, 2016.
- [21] J. P. Moore and M. D. Rogge, “Shape sensing using multi-core fiber optic cable and parametric curve solutions,” *Opt. Express*, vol. 20, pp. 2967–2973, 2012.
- [22] D. C. Rucker and R. J. Webster III, “Statics and dynamics of continuum robots with general tendon routing and external loading,” *IEEE Transactions on Robotics*, vol. 27, no. 6, pp. 1033–1044, 2011.
- [23] A. Gotelli, T. Zheng, A. Peloso, S. Teetaert, P. Dewi, G. Zetko, C. Shentu, and J. Burgner-Kahrs, “Multi-sensor dataset of a tendon-driven continuum robot in dynamic motion,” *figshare*, 2026. DOI: <https://doi.org/10.6084/m9.figshare.31990188>.